

ActInf Textbook Group ~ Cohort 1 ~ Meeting 12 (Review, Questions, Feedback)

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 12 (Review, Questions, Feedback) | 57:54

hey everyone

welcome to week

12

of

octave textbook group cohort one

we are in the review and synthesis

weeks

it's been a great

three-month-ish

journey

and it doesn't stop here but we'll go

over that so

two

points of process for those who are here

live and of course re-watching

and then we'll move to the more

conceptual parts

so first

in the onboarding page

you'll see these two columns the two

columns are rsvping yes to join cohort

two part one going through chapters one

through five if you would like to for a

second time

many people will be just doing it

for fun or for learning or however they

want and also there's the opportunity to

continue on with the second half of the

textbook

which is going to get into a lot of the

hands-on modeling

and we'll be especially looking forward

to developing a lot of the notebooks and

interactive formats that'll help us get

a lot of understanding there

um so please check
yes
for whichever
row your name is in for whether you
would like to continue on with part two
as well as
retake uh part one as well
you'd be welcome to be just a
participant
and also feel free to get in touch if
you are interested in taking some other
roles such as like facilitating or
scaffolding some aspect but we can
talk more about that for those who are
interested so
point one go to the onboarding list and
continue on
however you want
second point um if you go to future
textbook groups page
you'll see an embedded form
as well as a link to the form if you
want it in a new window like that
this would be
exceptionally requested and helpful
for you to provide some evaluations
numerically and as short or as long as
you want
on providing feedback
those are
one way that you can provide feedback
anonymously
the second way
is uh building on some of the ideas that
people are already adding here this is
an editable page so people can
add thoughts
as they see fit about like what would be

awesome for future textbook groups
or any other feedback or as always they
can email active inference
activeinference@gmail.com if they want
like a response to any specific points
um and then also it's provided here at
the link to share for people who want to
join future cohorts so that link will be
like getting people
onboarded into the next cohort which is
september 2022 in this case but
this is going to be like an evergreen
form
that will just continue to have a list
of people who expressed interest and be
on boarding them into
subsequent cohorts of which will begin
several per year
going forward
any
uh and next week we'll be talking more
about the feedback
specifically
and about projects
and about carrying on
today is like a little bit of a more
conceptual synthesis and review
and again next week will be more like
logistical
um and project oriented review so just
on these points or anything else
does anyone want to just raise their
hand or unmute and share anything that
they like
okay
let us
conceptually review
then

um

we read the first five chapters of the

active inference textbook

does anyone have overall thoughts that

they'd like to provide at the

five chapter scale

then we're going to go into a chapter

scale

then we'll be continuing to

dive in as um granularly as required

but at the scale of the first five

chapters you can see them here on this

preface page

as well as here in the chapters list

as a unit of five

chapters

what were people's sense

how did it update yes please ben

and then only

well i was just going to say i think um

you know as i've said to people before i

think

coming from a non-mathematics background

i thought that these chapters were

really accessible in terms of the kind

of conceptual

uh things that they were putting on the

table but i i just wondered what other

people's opinions were in terms of

um

you know

understanding the conceptual toolkit of

active inference

before you've built up a kind of

mathematical understanding

um because i'm about to start kind of um

learning some of the mathematics behind

it and i wondered what people thought of

like

the the potential to fully grasp these
concepts without the mathematics and
what the relationship between the two
might be i'm just curious what other
people think about that i guess
thanks great question um ali and then
mike and then anyone else who wants to
chime in

yeah it was a pretty exciting journey
for me uh but

i'll definitely need to
read over all those five chapters once
again at least once again uh in order to
grasp uh more fully uh the contents of
those chapters uh uh my own
personal opinion about uh
the way materials uh is organized in
these uh the first part of the textbook
is

uh

uh i couldn't see a kind of um
hierarchical or uh

i don't know structural uh
organization for these chapters as much
as i i like to see them because um
sometimes uh
they delve into a theoretical uh
theoretical aspect of all the things
much more deeply than the mathematical
side and sometimes it's vice versa
but

i i think that if it was
organized in a way that
as ben mentioned we can go from a
firm found a firm
conceptual foundation into
the granular formalism and mathematics

uh
at least for me that would be
much more uh accessible and much more uh
i would be able to organize the
materials in a coherent much more
coherent way in my mind so uh one thing
uh that um i really enjoyed about uh
these chapters is uh the

uh
kind of uh

uh
you see uh

a kind of uh the vision it uh

[Music]

tries to
put forward uh regarding the future uh
possible research uh and also uh things
especially in chapter five uh and the
the path that can be taken from this
point on

thank you

mike and then anyone else

yeah i'll pick up from where ali left
off i think it's

um
somewhat remarkable that they were able
to fit everything they did into those
first five chapters

and understanding

they probably had some
idea of how much space they wanted in
that part of the book before getting
into the the practical or applied

components in part two

you know the the high road low road
contrast and

um

sort of coming at the problem from

two different angles i think was useful
um
like ben i found the math to be
generally accessible although i confess
i i think i still go through this sort
of pattern where
i feel like i get it and i have this
intuitive understanding of it but then
there are things where
uh i sort of bog down and
looking at some of the mathematical
details or
some of the graphical representations um
related to things like message passing
um
have to slow down and
sort of take those apart more and
uh
i guess related to that i i should add i
think this group has been so good at
taking apart the ideas of the book this
is perhaps
one of the most effective settings i've
seen for
really
ringing
concepts out of a text and uh you know
the work that's going on with the coda
to build up the ontology and take apart
the equations and things like that i
think is just tremendous
awesome thank you like
yeah it's um
you know the british call them maths
plural
instead of math as an area and it really
does engage
like

there's visual formalisms which could be represented with a sparsity matrix or it could be represented with other um ways but there there are um almost like a multi-scale diversity of formalisms ranging from more traditional equal sign in the middle mathematics to fusion schematic equation graphics um different types of notations even within the equation um and the difficulty is not signaled not that it has to be but the difficulty does provide a little whiplash sometimes because it moves variously in the main thread of the text the boxes and the appendices from like this is how linear algebra works and this is what bayes theorem is all the way to topics that are approaching physics flows on partitioned states various kinds of fundamental or like in principle relationships about mathematics generally all the way to postulated

architectures

where message passing is implemented

to achieve some computational function

and or

thread the needle with resembling neural

computational architectures

the math is doing a lot

many things are happening and and so

it's um

only to be expected that the perception

is

challenging

and

there's probably a lot more to even

say

and um

unpack there

anyone else want to raise their hands

and

give a thought on this

section one area

if and then anyone else

have kind of a general question about

the limits

of mathematics

the notion of computational

irreducibility

uh

that

you have to

go through every step of a process

to get at state and plus 1 from

state and from state and minus whatever

that

certain problems cannot be

solved

by plugging values into a formula and

generating

a future state you have to compute all
the intermediate states
i wonder
where that
fits in in this picture
are there hard limits to the application
of of of mathematics
um to to to the problem of active
inference it's just a general question
returning to some um earlier
comments on
the firm conceptual ground
what
is an area no matter how
narrow or distal from actin where people
believe that
there is
a firm conceptual ground they felt like
that provided a foundation for them to
learn further
are we seeking an analogy
to some other
field of theory or practice or domain
or are we kind of seeking for a
epistemic
territory that we haven't quite
scene
i'll
just add i think that what in active
inference what you're talking about
there jeff is literally blankets
and uncertainty
like the corollary there of like
computational irreducibility
is that there's some large amount of
uncertainty in states in the world that
you're just never going to get at
the blanket is in some sense

not just

how you're relating to it

but some

hard

limit on your

you know your surprisal and

things like that so

thanks

uh brock had written oh yes mike please

uh so responding to the question about

maybe adjacent spacex or spaces or topic

areas

uh i think there's certainly overlap

with

concepts from system dynamics and

certainly a lot of agent-based modeling

has been applied

uh

to model systems that have feedback

loops and

adaptation things like that

we discussed

during the the past weeks relationships

with reinforcement learning and

um

i think there's

a belief that active inference kind of

cleans up some of the challenging

or maybe not challenging but uh

less formalized or less defined aspects

related to reinforcement learning

such as uh how do you engineer a utility

function that makes sense in the context

of what you're trying to model

banks agreed about creating a uh

integrated utility function

there's so many

unprincipled not to say ineffective or

not even to say not elegant but
unprincipled methods of creating
a chimera utility model
like
delay discounting
novelty bonus curiosity bonuses
alternating phases
or hyper parameters
that
um as well as purely implementational
strategies like
discarding burn in
parallel chains there's like
a whole toolkit and indeed fields
on creating
effectively
integrated utility models
and
putting that work
into the construction of the generative
model
and how the generative model is
partitioned from the generative process
allows
a generic free energy functional or set
of related free energy formulations
to play that role
so that's one very interesting angle
as well as
the way that that process of specifying
the generative model and generative
processes and so on
are
world models
whether
in the john le
sense
or in the atom saffron sense

these generative models encompass world models and deep learning approaches and various topics like that um so let's see blue wrote i wish there was practice problems in the book to develop our understanding yes um to a large extent and also it's whether people want to reflect on any specific questions or even just on this um scheme these are many of these questions these questions can be um open infinite game type questions about the material they can be clearly addressable questions about the material they can be um checks for understanding but we'll be continuing to develop and improve and curate questions around the material so i [Music] think it's a great comment and this is why we have this future textbook groups and all of these affordances for people

to stay involved and and be improving
this as well as experimenting in their
own
spaces and ways
to to develop the kinds of material that
are helping them
learn and understand
any more section one
overview thoughts
chapters one through five
and the framing of the book
being that the first five chapters are
more focused on the conceptual
background
and the second five chapters
are going to be
starting with a recipe for designing the
active models and and getting more into
the modeling itself
so any
section one through five
commentary
the book comprises two parts page three
these are aimed at readers who want to
understand active inference
first part
and those who seek to use it for their
own research second part
the first part of the book introduces
active inference
both conceptually and formally
contextualizing it within current
theories of cognition
how
did they succeed
and where was there a divergence between
what you
preferred

a priori or now
comprehensive formal and self-contained
introduction to active inference
its main constructs and implications for
the study of brain and cognition
i'm uh i'm interested in the distinction
between uh
understanding active inference and using
it
for our own research
and the way that that's set up as those
being two completely separate things
um
it seems to suggest that one could fully
understand active inference and yet not
be in a position to use it in their own
research
not having worked through the second
half of the book and i think that's
quite interesting
it is
good comment thank you
um
[Music]
not to uh spoil the ending
but the last sentence is ultimately we
are confident that you will continue to
pursue active inference in some form
there may be a multi-year
incubation
for different people and contexts
many people
now or even in some
amazing decentralized science future
they may not think of themselves as
researchers or of doing research
though they may be even included on
research projects

um

so it's it's very interesting uh

comments

about

how they

distinguish like

theory and learning

from

[Music]

part two

research and practice

however

um

[Music]

they

discuss applications

but this isn't

the playbook it's not the toolkit

it's not the modeling tool ecosystem

mike

yeah that's an interesting comment at

the end

and

i know this is my perceptual bias but

there seems to be this implicit

assumption that

active inference is the way to go

basically this is the tool to use

within the text

and

so as a result what's not there is a

discussion of

uh maybe these are cases where you might

not want to use active inference or

these are cases where applying active

entrance could create challenges in what

you're solving for

ali and anyone else

and in fact one of the uh most popular criticisms puts toward the active inference or fep in general is that it tries to explain everything it ends up explaining nothing so yeah i think there's a risk of maybe not sufficiently constraining the problem space or leaving things undefined in a way that uh allow for active interest to be the solution without a critical point of view they highlight behavior and cognition i mean one could almost see even the name active inference as being a synthesis of action behavior and cognition inference in ways that as we're discovering and and unpacking they're integrated in ways that other formalisms of perception as a type of cognition so we can just lump it there other frameworks of behavior and cognition [Music] have not integrated or have approached from a non-first principles

or maybe they do follow first principles
like the higher the impact factor the
better the paper
something like that that's the first
principle
um
no just kidding um
but
placing it within the framework of
cognition
while also
surfing on a wave
of
something like pan cognitivism or like
pan
[Music]
it computationalism
expand the scope
and that's even before
one starts to enter into the more recent
research
especially
where there is a highlight on system
persistence
not simply in terms of the resistance to
dissipation
entropically
but in a relational context as repeated
measurements like in the quantum work
so
it's like
cognition oh like brains like that
schema with a brain or with like a
person
well oh by cognition
it's any system
[Music]
with a blanket

that we're interacting with through
repeated measurement
is that cognition where did cognition go
brock
i am
i
kind of want to i guess echo all those
things that were just said about the
common criticism and this
where does it apply where does it not
apply but
also i guess
um all those things that you were just
listing like i i wonder
i always hear that from people that have
not um
engaged with the material
um
and from
my engagement with it it seems like
all of the examples in which it is used
it are
um
complex dynamical systems that don't
have great that have at best
mathematical approximations and have
almost no or literally no kind of
analytical
approaches
it just seems like a good way to model a
system of things of entities that are
interacting
um
and i don't i i guess what i'm
to to put it into like one question like
what is uh is that kind of some
anthropomorphizing
or whatever projecting

of like

well it's applied to this it's applied

to that

it must apply to everything and um you

know how can it apply to all those

things how can it apply everywhere it's

like it's not applying to everywhere

it's just applying to these

complex systems which are

everywhere

but also don't have and you know we're

not trying to model the apple falling

from the tree or the hyper you know the

these sorts of simple cases or something

it's not being applied there

i mean is it i don't i haven't seen

actually you know now that i think about

it any

of those sorts of situations or it's

applied

to things that we already understand in

this like maybe in some back-checking

way but not in a like way that's

suggested that we should use this much

more complex

you know approach to something that's

kind of

check already got that

you know

um

thanks brock um

ali i kind of feel that uh some of these

criticisms uh

stems from some from the fact from a

serious

a lack of deep understanding of uh

concepts related to fap and active

inference

for example uh i came across
a phd thesis just today
uh
in which
it tries to
critique
everything related to mark of blanket
but
as i was glancing over it uh
i
i observed some serious
uh gaps uh in our argument throughout
all and all the whole the thesis and
i think uh materials such as this book
uh can definitely help in filling up
these gaps of understanding
which even
the
the serious researchers
are suffering from
yes thanks
so to um
this thread
i've never heard it about the linear
model
how could the linear model be used in so
many settings
it explains nothing
well is the linear model an explanation
or is it an investigative tool
in the investigators toolbox
so
at least for me personally doesn't need
to have
broader reach than those who
resonate with it but
again i always try to ask
would somebody say this or could they

say that about a linear model
and then if the actin formalism
with
f equals
dot or g equals
dot
and all of the predicates that it's
associated with and the partition and
all of that
could someone say that about y equals mx
plus b
and then if not and there may be cases
where it is not the same
what is it that's different about active
inference
or
is it
a valid and important and even
reasonable question
but one that might be bumping up like
what jf raised about computational
irreducibility
or about map territory
even
disguised or camouflaged questions about
relationality
well how can complexity science apply to
so many topics
we're applying complexity science to
whatever topic we want to
where's the issue
mike and then ron
yeah a lot of it hinges on
how the model is to be applied and what
the intent of
model application is so
taking the linear model as an example we
might use something like that

to be predictive about a linear system
and so there's potential value there
uh when we do system dynamics models
a lot of times the value in doing a
system dynamics model is aligning the
stakeholders around
what are the model components and how do
we think about those model components so
working through a process similar to
like what we've done in this textbook
group although few do it as rigorously
to
identify the system elements and how
they interact and interrelate and so
forth and and that can lead to
more advanced implementation like
agent-based models
um
but a lot of times at least in my
experience those agent-based models are
used for
deeper systems understanding more than
say predictive power for
what the system is going to do at some
point in the future so
again it comes back to the
the reason for applying the models and
thinking about how the model fits with
that reason and what sort of results you
would expect from that
thank you mike ron
uh yeah i i generally agree with
what you and mike said
so from a practitioner's perspective
uh
i mean we do hear criticisms of
linear models right it's not that we
don't

but it's usually about which domain is
being applied in
uh
yeah uh
but generally speaking
well uh let me close this with uh
because active inference is probably
more
complex and it takes a lot more time to
drop than $y = mx + b$ which could
be
one reason for these kind of
criticisms and it's probably
like that they're falling back into old
habits okay why should i jump my old
model
as opposed to adopting this newer more
uh
well how much more does it explain than
what i hope
i think that
probably is driving a lot of the
criticism
of these things
thank you
i can't help but add a domain specific
example
um this is a 2016 paper
that is speaking to like a multi-decade
multi-career
brouhaha
about
different approaches to modeling
evolution and selection
for example in the eu social insects and
in social animals as well
and this article is very fascinating
because

they use a bayesian causal graph
to identify situations where
two different formulations
the kin and the multi-level selection
models
have identical predictions
where the causal graphs have identical
predictions
any measurement
you can think of as being like on a y
equals x manifold where it doesn't
resolve your uncertainty about which
framework is correct or not
yet the literature is littered with we
measured these ovaries
so can selection is or isn't happening
however a vast set of those empirical
cases
might be
essentially falling on this manifold
where those two models are not
distinguishable
and so
this is a theory driven approach like a
first principles approach
to identify situations
that
are
providing unique informative value
in this case about
canon multi-level selection
and knowing what territories
measurements will not have explanatory
value
which
relates to active inference um
just like rohan and like many people
have brought up especially when we

kind of take like a meta science or like
a communicating science or onboarding
people into active concern
like people are asking
explicitly or implicitly why should i
update my cognitive model
what is the value of active inference

can you

appetize me

with a two minute video

so i can understand it

or with a two minute video so it's

enough to want to continue going down

that path

there's a lot to say and there's um

[Music]

a lot of work

to identify

the situations

and the ways of truly addressing and

perhaps even resolving

[Music]

long-standing

[Music]

scientific divergences

like

if it is the case that the explore

exploit dialectic

is addressed in a novel way

through the free energy functional

balancing pragmatic and epistemic reward

that is

quite a vast scope

people continue to use explore and

exploit today

and it is the evident that that those

ontology terms wouldn't be useful in the

future

in fact
some of the
um
discussions on like folk psychology with
the belief desires intentions
can we say what an active inference
entity wants
or intends based upon its beliefs and
its desires for example
can we still talk about exploratory and
exploitative behavior
as a phenomena but have a different way
of modeling how that behavior arises or
or is
regulated
what are the settings in
behavioral cognitive science
what are the settings in
where systems dynamics has been applied
where we can now use active
inference to identify
where have we been just
throwing a thousand darts
at
the same
grain of sand
where are the vast
fields
where we don't even know
and as people have highlighted the book
is written
with a specific rhetorical
bent both explicitly and implicitly
um
for the regime of attention of the book
to be about active inference
one can read as an implicit endorsements
that this is something that is

valid or valuable to pursue
and
especially at this stage in the field it
is almost like an exhortation
to
persist amidst uncertainty
to learn and apply
few disjointed thoughts there but
i hope it makes sense
let's look back to the subjections of
the book
anyone can raise their hand or give a
thought like
how
could
the book you write
begin
what reviewer comments
whether you're reviewer one reviewer two
reviewer three
what comments would you have provided if
this were like
a draft
and you could even make structural
suggestions
so i have a lot of comments there
yes blue and then ron
um i uh
so like
having taught
lots of classes before like i think um
a list of like key terms and definitions
would be like critical i mean i mean and
like
learning objectives for each chapter
um
like i said uh like a practice quiz like
test your knowledge of this like like

what is surprise like how many different forms of surprise are there you know i mean like like just some practice questions to see if you grasped the concepts prevented in each chapter um

and especially like uh what i was mentioning in the chat like math problems like practice like can you set up your own generative model um like like what would the like or even just the bayesian graph like can you make a bayesian graph like what would that look like for you know give a give a question like a verbal description and then give the bayesian graph like answer or like make the answers available online so you could test your own knowledge of the subject first um i think that that would be like totally instrumental in in using this functionally like a textbook anyway so sorry

thanks blue nice predictive programming rohan and then

brock uh yeah uh so

my

as a practitioner right i work in the engineering field i'm a control systems engineer my preference would be to immediately see some application not necessarily to control systems but something that it begins with like a project and explains how active inference is better than its own baseline that would be interesting to see

uh like a perfect example of this would be

i don't know if you've heard of fastai by rachel thomas and jeremy howard they have a very unique way of teaching uh

so they have a computational linear algebra course so the way they teach the linear algebra part is by basically implementing gaussian blur for example and then connecting it to so you start with the code and then you do the math then you connect it to the math

so something like that might be far more useful

uh maybe not from a point but it would

definitely be useful for practitioners around the world okay so this

so if you know controls engineer gets this okay this looks like a better form model predictive control i can replace my model predictive controller with something

that resembles active inference

it might actually make uh

we might actually make more progress

that it actually gets out of the way

thank you ron brock

um

yeah i i think i mean i've had

similar comments to that of um

you know

fast ai and

benchmarks that that's kind of the ultimate

way that's just the way that um

practitioners change their um attention

now

um and that would be really useful but

in in the context of the book

um

i think i agree with every everything

else that was said too uh

i'm not sure how realistic it is to do

that from

the point you know where

the whole

field is at right now sorry about that

um

but

you know

i think

just having some some things in context

like there's always this comment about

like

you know it's a non-linear thing a text

is linear so of course it'll be out of

out of order

but like um

often the definitions of things are

kind of scattered throughout the

explanation of them

and

kind of

italicize instead of kind of brought up

to the top you know

in some

um

way you might expect

that i at least from my

my

pedagogical experience everything like

all math is

taught to you know

like to define your knowns and unknowns
and variables at the top and the sort of
thing right um
the in the
context of like the appendices
um
again there's like
i understand that
it's a lot you can't you know do you
can't add that all in or whatever but
just
if there was like
um either every whenever
the next part of the appendice became
relevant
you know um again just like having it at
the top like at each section you're like
okay so we're gonna use part you know
now building on you know what we've done
already like now we're going to use this
part of this this section in the
appendices in this section our equation
one two three whatever
in
in in this next section like just
stating it up front
again just like a like a you know
big box on the top or whatever
um
it just
would help like in wayfinding and
disentangling
the
unfortunate linear um
format of books
that is mapped to this complex thing um
and so
[Music]

yeah i don't know those those general things would be helpful one other thing that is maybe beyond the scope of this book but um spoke to ellie about very briefly at one point about the notation like even in the basic example of the frog here um like bayesian notation it's um it's incredibly easy to manipulate variables when they're one letter and subscripts but when the whole variable is like 10 or 20 letters like you know frog p p of x equals frog jump you know condition on a y it's like it becomes a larger i don't i don't know if this is just me and dyslexic you know like i don't know but um it just seems harder to cognitively kind of um work with that um i don't know thank you brock yes the chunking is multi-level and there it will be great to explore ways to um learn around that here are a few forays where we can copy quotes and have inline definitions so for those who are interested in this kind of work with an open source textbook it is not an infinite task or even a

non-automatable task
to copy out the plain text
and integrate it with a versioning
ontology
so that the definition
can be like
even rendered differently it could be in
a different language it could be every
single time that a term comes up
there's like
so much enrichment
that this seed
can lead to
of the textbook
and
a lot of it is for
us to pick up
honestly
rohan and then anyone else
oh i i forgot to put down
oh no worries thank you
so in our um
last seven or so minutes for this
session
next week
we will continue with any
conceptual points and questions that
people want to raise
we'll also be
turning our attention
a bit more
to
uh making sure first off that please
please everyone complete the feedback
form
in the future textbook groups and or
add information here or um contact us
because

it's just one of the most important
feedback mechanisms we have
we'll also be exploring a little bit
different
project
ideas
um
several of these threads involving
different system models
we've already found
a home for
in the active block prints project
where we're
meeting
twice on wednesdays weekly
other
projects
we're going to have new vistas on during
and after
going through the second half of the
textbook
which
is uh still black and white static pdfs
but we'll be able to augment this
heavily
with some
simulation tools
as well as
other
work like
the plain text enrichment or even anyone
who wants to collaborate on an audiobook
version
there's um
no lack of
act imp
tasks
today

and these are like
high leverage point
moments
where staying in the game and improving
the game will do tremendous service
towards reducing research debt and
improving the rigor and the applications
that we see

rohan

yeah uh can i make a project suggestion
that's fine yes please

everyone can yeah so that

yeah okay so there is uh i'm gonna put
this in the chat there's this online lab

by georgia tech's intelligence

assistance unit

uh called robutorio

uh that's basically a bunch of swarm

it's to test swarm controllers

so uh there's a good baseline over here

it would be a good idea to try like an

active inference version

like an active inference inspired swarm
controller

to finish the tasks over here

so this runs on actual robots

in that lab

and you can see the results

this is awesome um

i had

some

colleagues who worked with

some of the ant groups here

it's right at the intersection of

like

jf's work and many's interest in

embodied and robotic angle

and we have

somewhat

extensive

um

multi-scale generative models

of ants

which would apply to other swarm

settings as well so absolutely

um

we

want to scaffold those projects and

one way to start but not the only way to

start

is to

copy a version of this

these fields and add and remove as

relevant for you

and give some

um

handle

for others to be involved or just

stay in the game

so that um

more people can join there will be a

flow of people entering active inference

forever

how will we greet them

so uh what would i need to put in

um

like

it depends on how you want to carry

forward the project but you can feel in

this row add any

notes you'd like

and um

more details on what you're thinking of

doing

and then however you want to be more

specific about like

directions
and
and
just signal it in the idea
or you can in the um notes or or draft
catechism
which is what these templates are
based on
and if there is ever something you're
not
sure about then
ask around and um
different projects are going to
proceed
differently
okay yeah uh so i'll fill that out
with my talk
so awesome
well
interesting meeting
it's our penultimate discussion in this
section
then we hope that
everybody who is
motivated and
ready to do so
continues for part two
and or rejoins for part one cohort two
both of these will be starting in
september
so we'll take august
um off of the textbook
and
uh at this point the tentative plan is
that we'll have
two sequential hours
like
section one

and then followed by section two
so that those who are on section one
can um
see how
section two is
we can explore different architectures
and stuff and again this is like why
people's active participation and their
feedback
and the over sharing
on topic
and just writing all the questions that
came in their mind and all the things
that they would want to ask somebody to
check like
every contribution is important
does anyone else have any
thing they'd like to add before we stop
the recording
okay
stopping the recording
oh yeah someone go for it
all right stopping the recording